

Prehistory of AI

1931

The Austrian Kurt Goedel shows that in first-order predicate logic all true statements are derivable. In higher order logics, on the other hand, there are true statements that are unprovable.

1937

Alan Turing points out the limits of intelligent machines with the halting problem.

1943

McCulloch and Pitts model neural networks and make the connection to propositional logic.

1950

Alan Turing defines machine intelligence with the Turing test and writes about learning machines and genetic algorithms

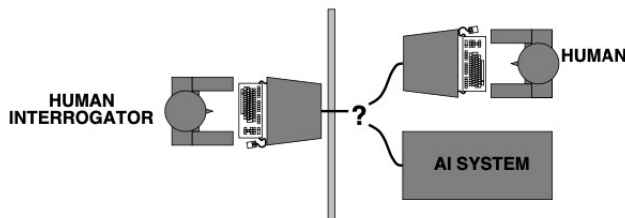
1951

Marvin Minsky develops a neural network machine. With 3000 vacuum tubes he simulates 40 neurons

Turing Test

Turing (1950) “Computing machinery and intelligence”:

- ◇ “Can machines think?” → “Can machines behave intelligently?”
- ◇ Operational test for intelligent behavior: the Imitation Game



- ◇ Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- ◇ Anticipated all major arguments against AI in following 50 years
- ◇ Suggested major components of AI: knowledge, reasoning, language understanding, learning

Problem: Turing test is not reproducible, constructive, or amenable to mathematical analysis

AI's official birth: Dartmouth, 1956



“An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. ***We think that a significant advance can be made if we work on it together for a summer.***”

**John McCarthy and Claude Shannon
Dartmouth Workshop Proposal**

Shakey the Robot (1966-1972)

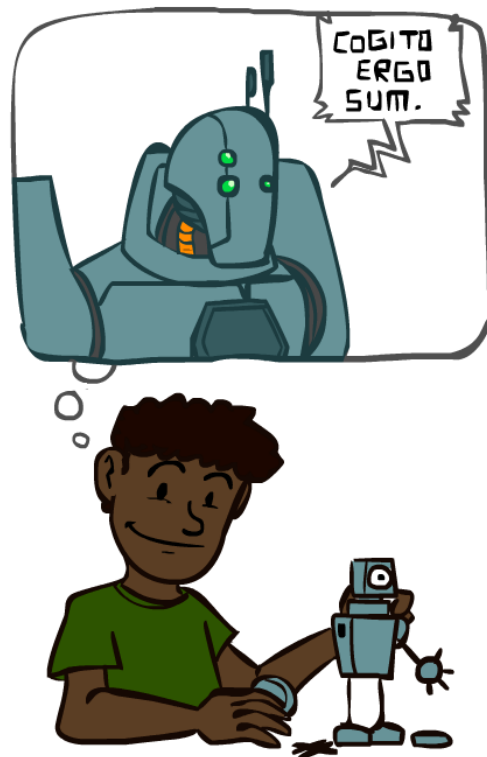
- Short video on Shakey
 - <https://www.youtube.com/watch?v=7bsEN8mwUB8>
- Long video on Shakey
 - <https://www.youtube.com/watch?v=GmU7SimFkpU>

Wiki: https://en.wikipedia.org/wiki/Shakey_the_robot



A (Short) History of AI

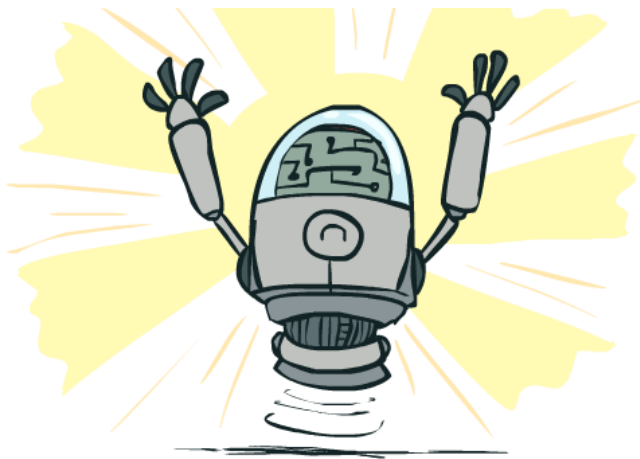
- 1940-1950: Early days
 - 1943: McCulloch & Pitts: Boolean circuit model of brain
 - 1950: Turing's "Computing Machinery and Intelligence"
- 1950—70: Excitement: Look, Ma, no hands!
 - 1950s: Early AI programs: chess, checkers (RL), theorem proving
 - 1956: **Dartmouth meeting: "Artificial Intelligence" adopted**
 - 1965: Robinson's complete algorithm for logical reasoning
- 1970—90: Knowledge-based approaches
 - 1969—79: Early development of knowledge-based systems
 - 1980—88: Expert systems industry booms
 - 1988—93: Expert systems industry busts: "AI Winter"
- 1990— 2012: Statistical approaches + subfield expertise
 - Resurgence of probability, focus on uncertainty
 - **General increase in technical depth**
 - Agents and learning systems... "AI Spring"?
- 2012— ____: Excitement: Look, Ma, no hands again?
 - Big data, big compute, deep learning
 - **AI used in many industries**



What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Play a decent game of table tennis?
- ✓ Play a decent game of Jeopardy?
- ✓ Drive safely along a curving mountain road?
- ✗ Drive safely along Telegraph Avenue?
- ✓ Buy a week's worth of groceries on the web?
- ✗ Buy a week's worth of groceries at Berkeley Bowl?
- ? Discover and prove a new mathematical theorem?
- ✗ Converse successfully with another person for an hour?
- ? Perform a surgical operation?
- ✓ Translate spoken Chinese into spoken English in real time?
- ? Fold the laundry and put away the dishes?
- ✗ Write an intentionally funny story?



Unintentionally Funny Stories

Once upon a time
and a vain crow
in his tree, hold
mouth. He not
piece of cheese
swallowed the
the crow. The



Janelle Shane

@JanelleCShane

Follow



Tried retraining the neural net on just "what do you get when you cross a X with a X?" jokes. Results did not improve. And for some reason, bungees are its favorite thing.

What do you get when you cross a dog and a vampire?

A bungee

What do you get when you cross a cow with a rhino?

A bungee with a dog

What do you get when you cross a street and a cow?

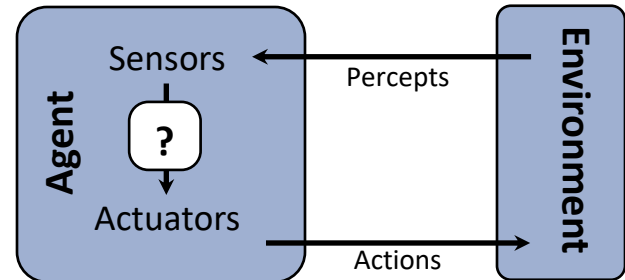
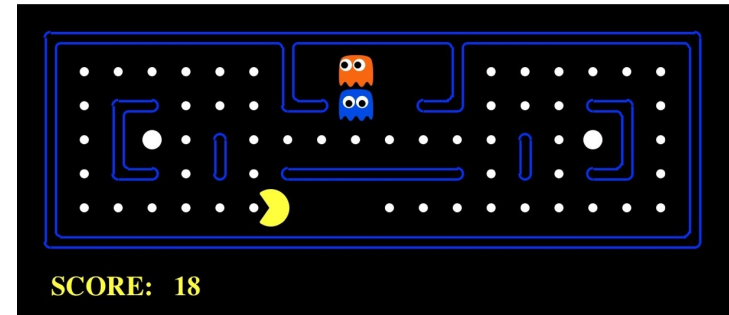
A bungee with a bungee and a rhino

What do you get when you cross a pig with a cow with a party?

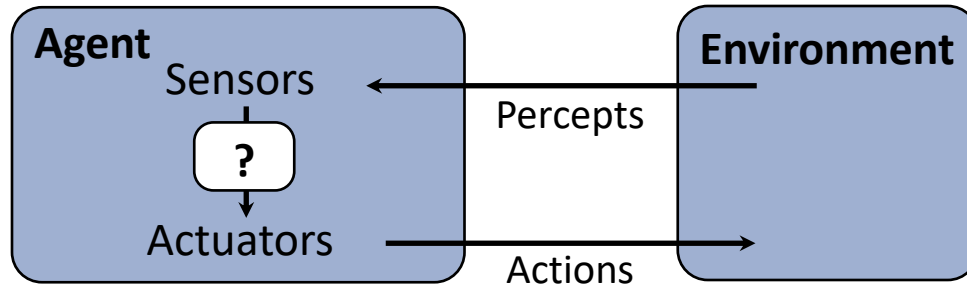
Because the engineers with a dog

AI as Designing Rational Agents

- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its expected utility.
- Characteristics of the **sensors, actuators, and environment** dictate techniques for selecting rational actions
- **This course** is about:
 - General AI techniques for many problem types
 - Learning to choose and apply the technique appropriate for each problem



Agents and environments



- An agent **perceives** its environment through **sensors** and **acts** upon it through **actuators** (or **effectors**, depending on whom you ask)
- The **agent function** maps percept sequences to actions
- It is generated by an **agent program** running on a **machine**

Rational agents

An agent is an entity that perceives and acts

This course is about designing rational agents

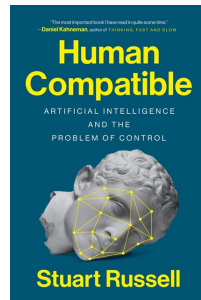
Abstractly, an agent is a function from percept histories to actions:

$$f : \mathcal{P}^* \rightarrow \mathcal{A}$$

For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance

Caveat:

1. Computational **limitations** make perfect rationality unachievable
→ design best program for given resources
2. Herbert Simon's **satisficing behavior**: how to apply it to AI?
3. Stuart Russell: be careful **not to act as a fanatic!**

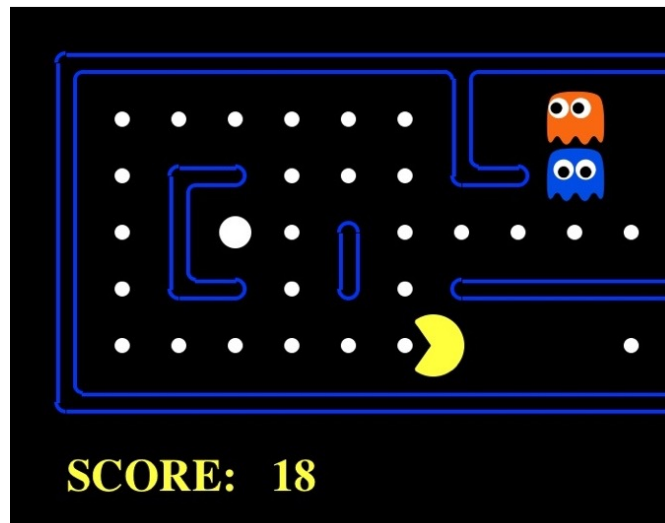


A human agent in Pacman



The task environment - PEAS

- Performance measure
 - -1 per step; + 10 food; +500 win; -500 die; +200 hit scared ghost
- Environment
 - Pacman dynamics (incl ghost behavior)
- Actuators
 - Left Right Up Down or NSEW
- Sensors
 - Entire state is visible (except power pellet duration)



PEAS: Automated taxi

- Performance measure
 - Income, happy customer, vehicle costs, fines, insurance premiums
- Environment
 - US streets, other drivers, customers, weather, police...
- Actuators
 - Steering, brake, gas, display/speaker
- Sensors
 - Camera, radar, accelerometer, engine sensors, microphone, GPS



Image: <http://nypost.com/2014/06/21/how-google-might-put-taxi-drivers-out-of-business/>

PEAS: Medical diagnosis system

- Performance measure
 - Patient health, cost, reputation
- Environment
 - Patients, medical staff, insurers, courts
- Actuators
 - Screen display, email
- Sensors
 - Keyboard/mouse

